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Exercise 2: C Compiler in Virtual Machine

Aim

To install a C compiler (GCC) in the Linux VM and execute simple programs.

Procedure

Step 1: Open Terminal

In the Ubuntu VM, press Ctrl+Alt+T to open the terminal.

Step 2: Update Package Repository

```
sudo apt update
```

Step 3: Install Build Essentials

The build-essential package includes the GCC compiler and other necessary tools.

```
sudo apt install build-essential -y
```

Step 4: Verify Installation

```
gcc --version
```

Step 5: Write a Simple C Program

Create a file named hello.c using a text editor like nano:

```
nano hello.c
```

Enter the following code:

```
#include <stdio.h>
int main() {
    printf("Hello, Virtual Cloud World!\n");
    return 0;
}
```

Step 6: Compile and Execute

```
gcc hello.c -o hello
./hello
```

Output should display: Hello, Virtual Cloud World!

Sample Program: Palindrome Check

A second program, palindrome.c, was written to check whether a given integer is a palindrome:

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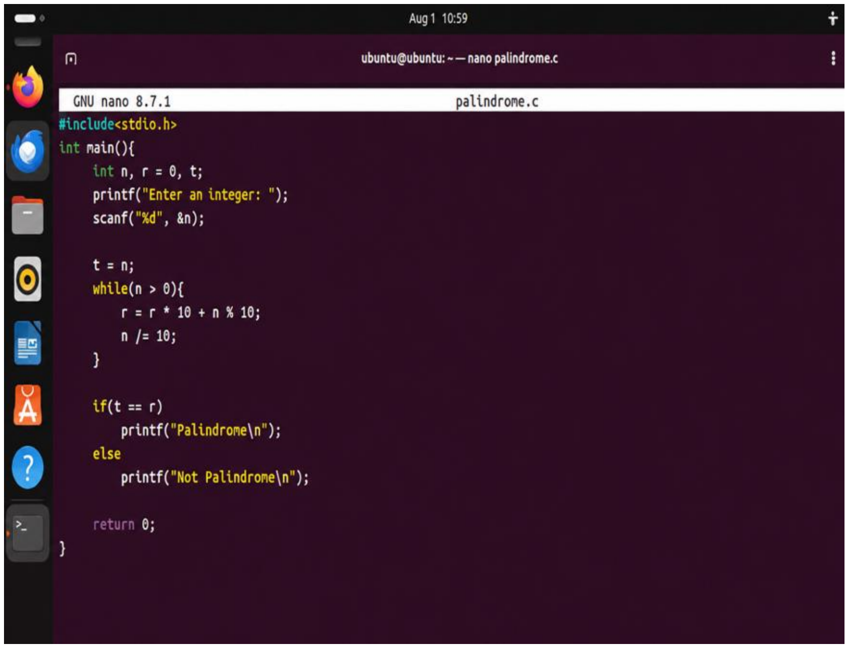
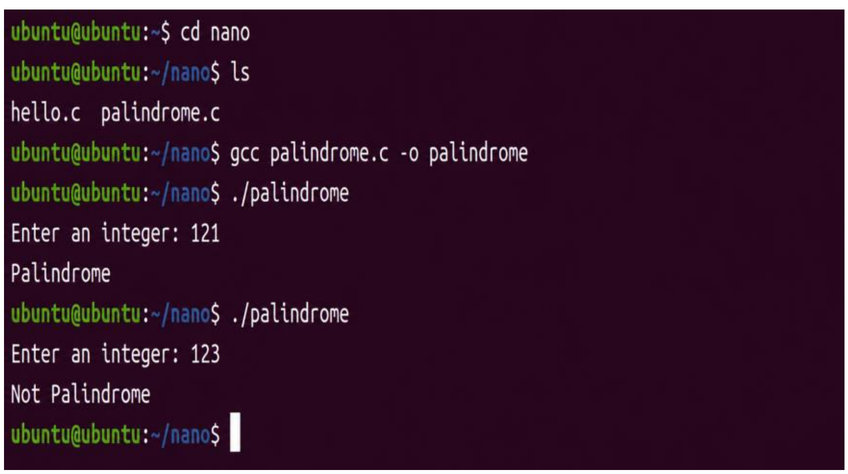



```c
#include<stdio.h>
int main(){
    int n, r = 0, t;
    printf("Enter an integer: ");
    scanf("%d", &n);

    t = n;
    while(n > 0){
        r = r * 10 + n % 10;
        n /= 10;
    }

    if(t == r)
        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");

    return 0;
}
```

Output

The palindrome.c program was compiled and executed, and tested with two inputs (121 and 123):



```
ubuntu@ubuntu:~$ cd nano
ubuntu@ubuntu:~/nano$ ls
hello.c  palindrome.c
ubuntu@ubuntu:~/nano$ gcc palindrome.c -o palindrome
ubuntu@ubuntu:~/nano$ ./palindrome
Enter an integer: 121
Palindrome
ubuntu@ubuntu:~/nano$ ./palindrome
Enter an integer: 123
Not Palindrome
ubuntu@ubuntu:~/nano$
```

Result

Thus, the GCC compiler was successfully installed in the Linux VM and used to compile and run C programs.